# A Pair-Resolved Scalar Morphology Probe of the White et al. Sphere–Cylinder Casimir Geometry

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May 29, 2026

**Abstract**

White et al. reported that custom Casimir boundary geometries can produce negative vacuum-energy morphologies with an unanticipated visual intersection with Alcubierre warp-metric source requirements. This paper reports a focused independent scalar probe of the sphere-in-cylinder portion of that report. The probe uses a role-resolved scalar morphology analysis: the sphere, cylinder wall, candidate shell region, readout axis, boundary infrastructure, and far-field control are kept distinct before the scalar field is interpreted. The interaction kernel uses paper-style v-loops and a pair-resolved segment/surface crossing rule: a loop contributes when its line segments cross both the sphere surface and the inner cylinder wall surface. Scale channels are reported separately, followed by a broad-window comparison.

In the mid-scale 0.75–1.50 $\mu$m window, the baseline sphere-cylinder geometry places 0.647–0.726 of the negative proxy magnitude in the near-shell band, with 9.8–14.2 fold enrichment over the rest of the sampled section, negligible far-field magnitude, and a stable radial peak near 1.125 $\mu$m from the sphere center. In that same channel, the explicit readout path carries 0.000651–0.001225 of the negative proxy magnitude. Modest sphere- and cylinder-size perturbations preserve the near-shell interaction pattern. Broad 0.25–5.00 $\mu$m integration becomes apparatus-wide in the scalar proxy and retains shell-enriched largest magnitudes. The calculation is a normalized scalar morphology probe; stress-tensor channels, SI normalization, material response, and metric-source closure belong to later calculation layers.

## 1. QUESTION

White et al. applied worldline numerics to custom Casimir geometries and found a negative energy-density morphology that they described as intersecting with the energy-density requirements of the Alcubierre metric [1]. The most compact version of that reported geometry is the sphere-in-cylinder toy model: a 1 $\mu$m sphere centered in a 4 $\mu$m cylinder. The reported calculation produced a shell-like negative distribution that was visually close to a two-dimensional section of an Alcubierre-style negative-energy shell.

This study evaluates two linked scalar questions:

> How robustly does the White et al. sphere-cylinder geometry organize negative scalar support into the proposed shell, and how strongly does that support couple to the central readout path under scale, seed, wall, and geometry variations?

Here "robust" means concentration in the intended sphere-cylinder shell region under seed changes, wall-thickness changes, separated scale windows, and modest sphere and cylinder perturbations. The

Alcubierre analogy depends on localized shell organization, with negative sign as one component of the evidence.

The analysis protocol draws on prior source-placement and role-separation work [4, 5]. The White device is treated here in its own Casimir terms: roles, channels, controls, and observables are separated before source or transport language is assigned.

## 2. GEOMETRY AND ROLES

The evaluated geometry is the White et al. sphere-cylinder toy model:

$$d_s = 1.0\,\mu\mathrm{m}, \qquad d_c = 4.0\,\mu\mathrm{m}, \tag{1}$$

with the sphere centered in the cylinder. The analysis treats the cylinder wall as boundary infrastructure, the sphere as the stress-shaping body, the annular region outside the sphere and inside the cylinder as the source-shell candidate, the central axis as the proposed transit/readout channel, and the exterior region as far-field control.

For a grid point $x = (x, y, z)$ in a two-dimensional section, let $r_s(x)$ be the distance from the sphere center and let $r_\perp(x) = \sqrt{y^2 + z^2}$ be the transverse radius relative to the cylinder axis. The near-shell band used for the principal concentration metric is

$$\mathcal{S}_{\mathrm{shell}} = \{x :\ R_s < r_s(x) \le R_s + 0.75\,\mu\mathrm{m}, \quad r_\perp(x) < R_c - 0.10\,\mu\mathrm{m}\}. \tag{2}$$

This band is a predefined role-labeled analysis region used for all reported cases to represent the intended sphere-cylinder shell, with the material wall and far field tracked as separate roles.

## 3. LOOP CONSTRUCTION AND KERNEL

The loop generator follows the v-loop construction quoted in White et al. The sampled Gaussian variables use the paper's $\exp(-w_i^2)$ convention, i.e. a normal distribution with variance 1/2. Let $y_{a,n} \in \mathbb{R}^3$ denote point $n$ on loop $a$, with the loop center of mass subtracted.

The scalar proxy uses segment/surface crossings. For each translated and scaled loop

$$\Gamma_{a,\lambda,c} = \{c + \lambda y_{a,n}\}_{n=1}^{N}, \tag{3}$$

the kernel determines whether any segment of the loop crosses the sphere surface and whether any segment crosses the inner cylinder wall. A loop contributes to the pair interaction when both crossings occur:

$$I_{a,\lambda}(c) = \mathbf{1}\{\Gamma_{a,\lambda,c} \cap \partial B_s \neq \varnothing\}\mathbf{1}\{\Gamma_{a,\lambda,c} \cap \partial C_{\mathrm{inner}} \neq \varnothing\}. \tag{4}$$

The normalized scalar morphology proxy at grid center $c$ is then

$$\mathcal{E}_{\mathrm{proxy}}(c) = -\frac{1}{N_{\mathrm{loop}}} \sum_{\lambda \in \Lambda} \frac{1}{\lambda^4} \sum_{a=1}^{N_{\mathrm{loop}}} I_{a,\lambda}(c). \tag{5}$$

The $\lambda^{-4}$ weighting keeps the scalar proxy aligned with the expected scale sensitivity of a four-dimensional Casimir-type contribution. The output category is a normalized scalar morphology proxy with explicit unnormalized-amplitude status. An SI energy-density estimate would require an additional normalization stage.

The single-body controls are embedded in the scoring rule. Contributions are accepted when a loop crosses both the sphere and the inner cylinder surface. This makes the probe a direct pair-interaction read; generic boundary contacts carry zero contribution in this score.

## 4. READOUT METRICS

Let

$$M_{-}(c) = \max[-\mathcal{E}_{\text{proxy}}(c), 0] \tag{6}$$

be the nonnegative magnitude of the negative proxy. The principal shell concentration metric is

$$F_{\mathcal{S}_{\text{shell}}} = \frac{\sum_{c\in\mathcal{S}_{\text{shell}}} M_{-}(c)}{\sum_{c} M_{-}(c)}. \tag{7}$$

The shell enrichment is the ratio between mean negative magnitude in the near-shell band and mean negative magnitude outside the band:

$$Q_{\mathcal{S}_{\text{shell}}} = \frac{\langle M_{-}\rangle_{\mathcal{S}_{\text{shell}}}}{\langle M_{-}\rangle_{\mathbb{R}^2\setminus\mathcal{S}_{\text{shell}}}}. \tag{8}$$

The top-decile shell fraction records the fraction of the strongest negative pixels that fall in $\mathcal{S}_{\text{shell}}$. Boundary, far-field, and readout participation are computed as the corresponding magnitude fractions in their role-labeled regions. Two central-channel quantities are tracked: the explicit readout path from the geometry role label, with radius $0.05\,\mu$m, and a wider central-axis mask,

$$A_{0.15} = \{x:\ r_{\perp}(x) \le 0.15\,\mu\text{m}\}. \tag{9}$$

The explicit readout path is the stricter transport-channel proxy; $A_{0.15}$ records nearby central-axis participation on the grid.

## 5. COMPUTATION CONFIGURATION

The calculation used the configuration in Table 1. The computational design emphasizes the interaction kernel and role-separated scale windows. Loop count and grid size set the scope of the present scalar morphology calculation; a full reproduction of White et al.'s worldline computation would add a larger loop ensemble and material-normalized stress-tensor reconstruction.

Table 1: Pair-surface scalar-probe configuration.

| Item | Value |
|---|---|
| Kernel | Pair-resolved segment/surface crossing |
| Surfaces | Sphere surface and inner cylinder wall |
| Loop method | Paper-style v-loop |
| Baseline seeds | $1, 7, 17$ |
| Wall thicknesses | $0.2\,\mu$m, $0.4\,\mu$m |
| Baseline scale windows | 0.75–1.50 $\mu$m, 1.50–2.75 $\mu$m, 0.25–5.00 $\mu$m |
| Geometry perturbations | sphere $0.9\,\mu$m, sphere $1.1\,\mu$m, cylinder $3.8\,\mu$m, cylinder $4.2\,\mu$m |
| Perturbation seed | 1 |
| Loops | 96 |
| Points per loop | 192 |
| Grid | $37 \times 37$ over $\pm 2.5\,\mu$m |
| Workers | 4 |

## 6. BASELINE RESULTS

Table 2 gives the baseline aggregate over the three seeds. The mid-scale window is the most spatially selective morphology channel. It covers about 0.35–0.37 of the sampled pixels with negative proxy values, yet it places 0.647–0.726 of the negative magnitude in the near-shell band. Boundary and far-field fractions are negligible, and the peak radial bin remains at 1.125 $\mu$m.

Table 2: Baseline pair-surface scalar-probe read. Values are means over seeds $1, 7, 17$.

| Window | Wall | Neg. frac. | $F_{\mathcal{S}_{\mathrm{shell}}}$ | $Q_{\mathcal{S}_{\mathrm{shell}}}$ | Top-decile shell | Boundary | Far |
|---|---|---|---|---|---|---|---|
| 0.75–1.50 | 0.2 | 0.347699 | 0.646612 | 9.782 | 0.781228 | 0.000283 | 0.000000 |
| 0.75–1.50 | 0.4 | 0.365961 | 0.726420 | 14.206 | 0.883074 | 0.001319 | 0.000000 |
| 1.50–2.75 | 0.2 | 0.892866 | 0.540825 | 6.289 | 0.771857 | 0.012854 | 0.005076 |
| 1.50–2.75 | 0.4 | 0.893109 | 0.567666 | 7.012 | 0.875146 | 0.023294 | 0.000718 |
| 0.25–5.00 | 0.2 | 1.000000 | 0.504602 | 5.437 | 0.708029 | 0.020623 | 0.014162 |
| 0.25–5.00 | 0.4 | 1.000000 | 0.547679 | 6.464 | 0.790754 | 0.032290 | 0.002884 |

The upper-scale window is broader and fills most of the section, and its largest negative pixels remain shell-concentrated. The broad integration window gives apparatus-wide negative scalar participation because larger loops connect the sphere-wall system across much of the sampled section. Even in the broad field, approximately half of the negative magnitude remains in the near-shell band with 5–6 fold enrichment. These rows separate two scalar channels: apparatus-wide broad-window participation and localized mid-scale pair interaction.

Table 3 gives the corresponding central-channel read. In the mid-scale channel, the explicit readout path carries 0.000651–0.001225 of the negative proxy magnitude, and the wider central-axis mask carries 0.002918–0.005459. The upper-scale and broad channels add apparatus-wide central participation at the few-percent level on the explicit readout path.

Table 3: Baseline readout-axis participation. Values are means over seeds $1, 7, 17$.

| Window | Wall | Explicit readout path | Central axis $A_{0.15}$ |
|---|---|---|---|
| 0.75–1.50 | 0.2 | 0.000651 | 0.002918 |
| 0.75–1.50 | 0.4 | 0.001225 | 0.005459 |
| 1.50–2.75 | 0.2 | 0.019870 | 0.063433 |
| 1.50–2.75 | 0.4 | 0.023333 | 0.073679 |
| 0.25–5.00 | 0.2 | 0.023119 | 0.071958 |
| 0.25–5.00 | 0.4 | 0.023688 | 0.073302 |

## 7. GEOMETRY PERTURBATIONS

Table 4 summarizes the mid-scale perturbation cases. The tighter cylinder strengthens the near-shell concentration, the wider cylinder weakens it, and both sphere-size perturbations preserve a recognizable shell-localized response. The peak radial bin remains near 1.125 $\mu$m in the mid-scale cases.

The upper-scale perturbation cases retain a broader and still organized read: near-shell magnitude fractions range from 0.515 to 0.595, top-decile near-shell fractions range from 0.719 to 0.942, and far-field magnitude fractions stay below 0.006655. The perturbation rows follow the same qualitative

Table 4: Mid-scale perturbation read at seed 1.

| Geometry | Wall | Neg. frac. | $F_{\mathcal{S}_{\text{shell}}}$ | $Q_{\mathcal{S}_{\text{shell}}}$ | Top-decile shell |
|---|---|---|---|---|---|
| Cylinder 3.8 $\mu$m | 0.2 | 0.379839 | 0.718409 | 13.618 | 0.865385 |
| Cylinder 3.8 $\mu$m | 0.4 | 0.385683 | 0.782528 | 19.208 | 0.962264 |
| Cylinder 4.2 $\mu$m | 0.2 | 0.310446 | 0.539340 | 6.250 | 0.627907 |
| Cylinder 4.2 $\mu$m | 0.4 | 0.372535 | 0.632558 | 9.189 | 0.745098 |
| Sphere 0.9 $\mu$m | 0.2 | 0.344777 | 0.628413 | 9.658 | 0.708333 |
| Sphere 0.9 $\mu$m | 0.4 | 0.352082 | 0.719429 | 14.643 | 0.897959 |
| Sphere 1.1 $\mu$m | 0.2 | 0.395179 | 0.689049 | 10.860 | 0.818182 |
| Sphere 1.1 $\mu$m | 0.4 | 0.401753 | 0.762371 | 15.723 | 0.909091 |

picture as the baseline calculation: the mid-scale channel is localized, and larger scale channels add apparatus-wide participation while the largest magnitudes remain near the shell. Across the perturbation cases, the mid-scale explicit readout-path fraction stays in the 0.000000–0.002634 range, with the wider central-axis fraction in the 0.000000–0.008408 range. The upper-scale perturbation cases carry 0.017245–0.027398 on the explicit readout path and 0.055520–0.084184 on the wider central-axis mask.

## 8. INTERPRETATION

Within this normalized scalar morphology model, the White et al. sphere-cylinder geometry produces a stable near-shell pair interaction. The mid-scale channel supplies the most spatially discriminating read: negative proxy magnitude concentrates in the annular source band, far-field magnitude is small, boundary participation remains secondary, and the radial peak stays near the middle of the sphere-wall gap. The sign map is accompanied by localization, role, and scale placement.

The broad-window field describes a separate scale channel. Larger loops connect the apparatus over most of the sampled section, producing apparatus-wide negative scalar participation. The broad field retains a shell-enriched largest-magnitude component and combines localized sphere-wall morphology with long-scale apparatus connection. The Alcubierre-like comparison is carried by the separated mid-scale pair interaction, which localizes around the intended shell band.

The readout-axis metrics separate shell formation from transit-path coupling. In the clean mid-scale channel, most negative proxy magnitude is assigned to the near-shell band, and the explicit transit/readout path receives about one tenth of one percent of the magnitude. Larger scale channels add weak central participation as part of the broader apparatus field. The scalar morphology is concentrated where a source shell would be drawn, with the central path appearing mainly through apparatus-scale coupling.

This leaves the source-function question at the tensor and material level. The Alcubierre comparison requires stress components, sign placement, magnitude comparison, and metric source demand in addition to negative scalar support. A proposed transit-time or current/photon/electron readout also carries ordinary electromagnetic competition: capacitance, impedance, dispersion, material response, contact loading, patch potentials, surface roughness, and geometry-dependent leakage can all dominate a laboratory signal.

## 9. CODE AND DATA AVAILABILITY

The repository contains the calculation code, machine-readable case table, summary data, and field arrays under the White Casimir experiment directory: `white_casimir_source_function_audit/`.

## 10. CONCLUSION

The scalar probe separates two parts of the White et al. sphere-cylinder proposal. The geometry forms a shell-centered Casimir proxy in the role-resolved calculation: mid-scale pair crossings concentrate in the annular source region, the radial placement tracks the sphere-wall gap, far-field leakage stays small, and modest sphere/cylinder perturbations preserve the same organization. The pair-crossing rule gives the annulus a built-in role, so annular activity alone carries little information. The combined radial, scale-window, leakage, and perturbation behavior carries the morphology.

The warp/transport reading asks for more structure than this scalar proxy supplies. The mid-scale morphology channel belongs to the shell geometry; the central transit/readout path is weak there and enters mainly through broader apparatus-scale participation. The White question moves from whether the toy geometry can produce a structured scalar shell to whether that shell can be converted into a metric source with material and transport behavior. The former is present in this proxy; the latter requires stress-tensor reconstruction, SI normalization, material response, and a transport observable tied to the central channel.

## References

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